

Synopsis

ONLINE PURCHASE PREDICTOR

Problem Id: 21

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- **PROBLEM DESCRIPTION:**

With the rapid growth of e-commerce platforms, understanding customer purchasing behavior has become extremely important for businesses and online retailers. Large numbers of users visit shopping websites daily, browse products, click on different pages, add items to carts, and spend time exploring products, but only a limited number of users finally complete a purchase. Predicting whether a customer is likely to purchase a product based on their online browsing behavior can help companies improve marketing strategies, provide personalized recommendations, increase customer engagement, and maximize sales conversion rates.

The objective of this project is to develop a machine learning-based Online Purchase Prediction System that predicts whether a customer will make a purchase or not using behavioral and interaction-based features collected from online shopping activity. The dataset used in this project contains multiple customer behavior attributes such as:

- browsing time
- number of clicks
- pages visited
- add-to-cart count
- product category viewed
- bounce rate
- time per page
- session duration
- product views
- user type
- search keywords
- location
- wishlist activity

These features provide valuable insights into customer interest, browsing patterns, and purchasing intent.

The dataset also contains realistic characteristics of real-world e-commerce data such as:

- missing values
- duplicate records
- random noise
- outliers
- inconsistencies in measurements

Therefore, before building machine learning models, extensive data preprocessing is performed to improve data quality and ensure reliable model performance.

The preprocessing stage includes:

- handling missing values using imputation techniques
- removing duplicate records
- handling outliers using the IQR method
- converting categorical data into numerical form using One-Hot Encoding
- applying feature scaling using StandardScaler
- performing correlation-based feature selection

Feature selection is used to remove low-impact and irrelevant features and retain only the most important behavioral attributes influencing customer purchasing decisions.

After preprocessing, multiple machine learning classification algorithms are implemented and compared, including:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Classification
- Support Vector Machine (SVM)

To improve model performance, hyperparameter tuning using GridSearchCV is applied to all models. The models are then evaluated using multiple performance metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- Classification Report

The final best-fit model is selected based on overall performance, scalability, generalization ability, and balanced classification metrics.

The developed system can help e-commerce businesses:

- predict customer purchasing behavior
- improve personalized marketing
- optimize recommendation systems
- identify potential buyers

- enhance customer experience
- increase sales and conversion rates

Thus, this project demonstrates the practical application of machine learning techniques in solving real-world business problems related to customer behavior analysis and online purchase prediction.

• **SYSTEM WORKFLOW / ARCHITECTURE:**

The Online Purchase Behavior Classification System is designed to predict whether a customer will purchase a product or not based on their browsing and interaction behavior on an e-commerce platform. The system uses machine learning algorithms to analyze customer activity patterns and generate predictions using preprocessed behavioral data.

The workflow of the system consists of multiple stages, beginning from dataset collection and ending with final purchase prediction.

1. Dataset Collection

The first stage of the system involves collecting the online purchase behavior dataset. The dataset contains customer-related behavioral features such as:

- Browsing Time
- Number of Clicks
- Pages Visited
- Add to Cart Count
- Product Category Viewed
- Wishlist Add
- Bounce Rate
- Exit Pages
- Time Per Page
- User Type
- Location
- Search Keywords
- Session Duration
- Product Views
- Purchase (Target Variable)

The dataset also includes realistic characteristics such as:

- missing values
- duplicate records
- outliers
- random noise
- inconsistencies in measurements

These characteristics simulate real-world e-commerce environments.

2. Data Preprocessing

After collecting the dataset, preprocessing techniques are applied to improve the quality of data and make it suitable for machine learning models.

a) Handling Missing Values

Missing values present in numerical columns are replaced using median or mean values, while categorical missing values are replaced using the most frequent category.

b) Removing Duplicate Records

Duplicate rows are removed to prevent repeated information from affecting the model performance.

c) Outlier Handling

Outliers are handled using the Interquartile Range (IQR) method to reduce abnormal data influence.

d) Encoding Categorical Variables

Categorical features such as product category, user type, and location are converted into numerical form using One-Hot Encoding (`get_dummies()`).

e) Feature Scaling

Feature scaling is applied using `StandardScaler` to normalize the dataset and improve the performance of models such as SVM and KNN.

3. Exploratory Data Analysis (EDA)

EDA is performed to understand customer behavior patterns and relationships between variables.

The following analyses are performed:

- correlation analysis
- heatmaps
- boxplots

- histograms
- distribution analysis
- outlier visualization

Correlation heatmaps help identify highly related features and understand how each feature affects the Purchase target variable.

4. Feature Engineering and Feature Selection

After preprocessing, feature engineering and selection techniques are applied.

Feature Selection

Correlation-based feature selection is used to select only important features related to the Purchase column.

Features with very low correlation values are removed to:

- reduce model complexity
 - improve accuracy
 - reduce overfitting
 - improve training efficiency
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5. Train-Test Split

The processed dataset is divided into:

- Training Data (80%)
- Testing Data (20%)

The training dataset is used to train the machine learning models, while the testing dataset is used to evaluate model performance on unseen data.

6. Model Training

Multiple classification models are trained and compared, including:

a) Logistic Regression

Used as a baseline classification model due to its simplicity and scalability.

b) K-Nearest Neighbors (KNN)

Used to classify customer behavior based on nearest neighboring patterns.

c) Decision Tree Classification

Used to create rule-based decision structures for customer purchase prediction.

d) Support Vector Machine (SVM)

Used to find the optimal decision boundary between purchase and non-purchase classes.

7. Hyperparameter Tuning

To improve model performance, hyperparameter tuning is performed using GridSearchCV.

Different parameter combinations are tested for:

- Logistic Regression
- KNN
- Decision Tree
- SVM

The best parameters are selected based on evaluation metrics such as F1 Score and Accuracy.

8. Model Evaluation

The trained models are evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- Classification Report

The performance of all models is compared to identify the best-fit model for the problem statement.

9. Best Model Selection

After comparing all models, the best-performing model is selected based on:

- highest F1 score
- balanced precision and recall
- scalability
- computational efficiency
- generalization ability

For this project, the Tuned Logistic Regression / Tuned SVM model showed strong overall performance depending on the evaluation results.

10. Final Prediction System

The final stage of the system allows the user to manually input customer behavioral details such as browsing time, clicks, session duration, and product views.

The trained machine learning model processes the input and predicts:

1 → Customer Will Purchase

0 → Customer Will Not Purchase

This prediction system can help e-commerce platforms improve:

- targeted marketing
- recommendation systems
- customer engagement
- sales conversion strategies

• STEPS OF IMPLEMENTATION:

1. Data Collection:

The first phase involves gathering the dataset containing customer behavioral attributes. This includes tracking metrics such as:

Browsing time and session duration.

Number of clicks and product views.

Specific page visits during a shopping session.

2. Data Preprocessing:

Before the model can learn, the raw data must be cleaned and organized. Key tasks include:

- **Cleaning:** Handling missing values and removing duplicate or irrelevant records.
- **Feature Scaling:** Adjusting features so they reside within a similar range for better mathematical processing.
- **Feature Selection:** Using techniques to eliminate low-impact attributes and retain only the most influential variables

3. Model Selection

In this phase, multiple classification algorithms are selected and implemented to handle the binary classification task:

Purchase (1) vs No Purchase (0)

The selected machine learning models for this project are:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Classification
- Support Vector Machine (SVM)

These models were trained, tuned, evaluated, and compared using Accuracy, Precision, Recall, and F1 Score to identify the best-fit model for the Online Purchase Prediction System.

Justification of Selected Algorithms

In this project, Logistic Regression, KNN, Decision Tree, and SVM are selected because they represent different machine learning approaches and allow proper comparison of classification performance on customer behavioral data.

The dataset contains features such as:

- browsing time
- number of clicks
- pages visited
- session duration
- add-to-cart count
- product views
- bounce rate
- customer activity behavior

Since the problem is a binary classification problem, these algorithms are suitable for predicting whether a customer will purchase a product or not.

1. LOGISTIC REGRESSION

Logistic Regression is a supervised learning algorithm widely used for binary classification problems. Since the target variable in this project contains only two classes:

1 → Purchase

0 → No Purchase

Logistic Regression becomes a suitable and efficient choice.

It predicts the probability of purchase using a sigmoid function and provides clear decision boundaries between purchase and non-purchase classes.

Reasons for Selection:

- Works efficiently on large datasets
- Computationally fast
- Easy to interpret
- Less prone to overfitting
- Performs well after feature scaling and preprocessing
- Provides strong generalization capability

During implementation, hyperparameter tuning using GridSearchCV was also applied to improve model performance and optimize parameters such as:

- regularization strength (C)
- solver type
- class weights

The tuned Logistic Regression model achieved strong Accuracy and F1 Score, making it one of the best-performing models in the project.

2. K-NEAREST NEIGHBORS (KNN)

KNN is a distance-based supervised learning algorithm that classifies data points based on similarity with neighboring data points.

Unlike Logistic Regression, KNN does not assume any linear relationship between features and target variables. It is useful for identifying local patterns in customer behavior.

For example:

- customers with similar browsing behavior
- similar session durations
- similar click patterns

may show similar purchasing tendencies.

Reasons for Selection:

- Simple and easy to understand
- Effective for behavioral pattern recognition
- Captures non-linear relationships

- Useful for similarity-based classification

Since KNN depends heavily on distance calculations, feature scaling using StandardScaler was applied before training the model.

Hyperparameter tuning was also performed for:

- number of neighbors (K)
- distance metric
- weight strategy

Although KNN performed well, it required higher computational cost because the dataset contains a large number of records.

3. DECISION TREE CLASSIFICATION

Decision Tree is a rule-based supervised learning algorithm that creates decision structures using feature conditions.

The model splits the dataset into branches based on feature importance and creates classification rules for purchase prediction.

Reasons for Selection:

- Easy to visualize and interpret
- Handles both numerical and categorical data
- Captures non-linear decision boundaries
- Useful for understanding feature importance

Decision Tree was included in the project to compare rule-based learning with probability-based and distance-based learning methods.

Hyperparameter tuning was performed for:

- maximum depth
- splitting criteria
- minimum samples split
- minimum samples leaf

Although Decision Tree provided good interpretability, it showed higher risk of overfitting compared to Logistic Regression.

4. SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine is a powerful supervised learning algorithm used for classification tasks. It works by finding the optimal hyperplane that separates purchase and non-purchase classes.

SVM performs well on high-dimensional datasets and can model complex relationships using kernel functions.

Reasons for Selection:

- Strong classification capability
- Effective on high-dimensional encoded data
- Good generalization performance
- Handles complex decision boundaries

Feature scaling was necessary before applying SVM because the algorithm is sensitive to feature magnitudes.

Hyperparameter tuning was performed for:

- kernel type
- regularization parameter (C)
- gamma values

Although SVM achieved strong performance, it required higher computational time due to the large dataset size.

5. Hyperparameter Tuning

To improve model performance, hyperparameter tuning was performed using:

GridSearchCV

Different parameter combinations were tested for all models to obtain:

- higher Accuracy
- better Precision
- improved Recall
- higher F1 Score

This helped optimize the models and reduce overfitting.

6. Overall Justification

By using Logistic Regression, KNN, Decision Tree, and SVM together, the project compares four different machine learning approaches:

Model	Learning Approach
Logistic Regression	Probability-based learning
KNN	Distance-based learning
Decision Tree	Rule-based learning
SVM	Margin-based classification

This comparison provides a deeper understanding of model behavior and classification performance on online customer behavioral data.

After evaluating all models, the Tuned Logistic Regression / Tuned SVM model achieved the best overall performance based on Accuracy and F1 Score, making it the best-fit model for the Online Purchase Prediction System.

4. Training and Validation

The preprocessed data is fed into the selected algorithms. The model learns to associate specific browsing patterns (like high click counts) with the likelihood of a purchase. Validation is performed to ensure the model isn't just "memorizing" the data but actually learning patterns

5. Evaluation

The model's performance is tested using unseen data to ensure reliability. Success is measured using:

Accuracy, Precision, and F1-score to determine how well it predicts customer behavior.

6. Final Model

Once the performance meets the required standards, the Final Model is generated. This model can then be used by e-commerce platforms to personalize recommendations and improve marketing strategies in real-time.

- **Expected Modules/ Libraries to be used :**

- 1. NumPy:**

For numerical computations and handling arrays.

- 2. Pandas:**

For data manipulation, cleaning, and analysis.

3. Matplotlib:

For basic data visualization (graphs and plots).

4. Seaborn:

For advanced and attractive statistical visualizations.

5. Scikit-learn (sklearn):

For machine learning tasks:

- Data preprocessing (scaling, encoding):
- Feature selection
- Algorithms like Logistic Regression and KNN
- Model evaluation (accuracy, precision, F1-score)

6. SciPy (optional):

For additional scientific and statistical computations.

7. Warnings / OS (optional):

For handling system operations and suppressing unnecessary warnings.

• EXPECTED EVALUATION METRICS AND PLOTS TO BE USED:

To evaluate the performance of the proposed machine learning model for predicting online purchase behavior, several evaluation metrics and visualization techniques will be used. Since the problem is a binary classification task (purchase or no purchase), multiple metrics are considered to ensure a comprehensive assessment of the model.

REQUIRED TOPICS FROM THE SUBJECT

1. Data Preprocessing

Data preprocessing is one of the most important stages of the Online Purchase Prediction System. It involves cleaning, transforming, and preparing raw customer behavioral data before training machine learning models. Since real-world datasets often contain missing values, duplicate records, outliers, and inconsistent data, preprocessing helps improve data quality and model performance.

In this project, several preprocessing techniques were applied:

- Handling missing values using median and most-frequent value imputation
- Removing duplicate records
- Handling outliers using the IQR (Interquartile Range) method
- Converting categorical variables into numerical form using One-Hot Encoding

- Feature scaling using StandardScaler

These preprocessing steps improved model accuracy, reduced noise, and ensured that the machine learning algorithms worked efficiently on the dataset.

2. Feature Selection

Feature selection is the process of selecting the most important and relevant features from the dataset for model training. It helps remove unnecessary, weak, or low-impact variables that may negatively affect model performance.

In this project, correlation-based feature selection was used after preprocessing and encoding the dataset. Features with very low correlation values with the Purchase target column were removed.

The main advantages of feature selection are:

- reduced model complexity
- faster training
- improved model accuracy
- reduced overfitting
- better computational efficiency

The selected features were then used for training all classification models.

3. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. Since the objective of this project is to predict:

Purchase (1) or No Purchase (0)

Logistic Regression was selected as one of the primary models.

The algorithm predicts the probability of purchase using a sigmoid function and creates a decision boundary between the two classes.

In this project, Logistic Regression was:

- trained on the preprocessed dataset
- scaled using StandardScaler
- tuned using GridSearchCV for better performance

The model achieved strong Accuracy and F1 Score and showed excellent scalability and generalization ability for large datasets.

4. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies data based on similarity between neighboring data points. It predicts the class of a customer by identifying the nearest customer behavior patterns using distance calculations.

KNN was used in this project because customer purchasing behavior may contain non-linear patterns that can be captured using similarity-based learning.

The algorithm:

- calculates distances between data points
- identifies nearest neighbors
- predicts purchase behavior based on majority voting

Since KNN is distance-based, feature scaling was necessary before model training.

Hyperparameter tuning was also applied to optimize:

- number of neighbors (K)
- distance metric
- weighting strategy

KNN performed well on behavioral pattern recognition but required higher computational cost due to the large dataset size.

5. Decision Tree Classification

Decision Tree is a supervised learning algorithm that creates rule-based decision structures for classification. The model splits the dataset into branches based on feature conditions and predicts whether a customer will purchase or not.

In this project, Decision Tree was used to compare rule-based learning with other classification approaches.

Advantages of Decision Tree include:

- easy interpretation
- visualization capability
- handling both numerical and categorical features
- capturing non-linear relationships

Hyperparameter tuning was performed to optimize:

- maximum depth
- splitting criteria
- minimum samples split

- minimum samples leaf

Although Decision Tree provided good interpretability, it showed higher overfitting risk compared to Logistic Regression.

6. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm used for classification tasks. It works by identifying the optimal hyperplane that separates the purchase and non-purchase classes.

SVM was included in this project because it performs well on:

- high-dimensional data
- complex classification boundaries
- encoded datasets

Feature scaling was necessary before training the SVM model because the algorithm is highly sensitive to feature magnitudes.

Hyperparameter tuning was performed using GridSearchCV to optimize:

- kernel type
- regularization parameter (C)
- gamma values

The tuned SVM model achieved strong Accuracy and F1 Score but required higher computational time compared to Logistic Regression.

7. Model Evaluation Metrics

Model evaluation metrics are used to measure the performance of machine learning models and compare their prediction capabilities.

In this project, the following metrics were used:

Accuracy

Accuracy measures the overall correctness of the model.

Accuracy = Correct Predictions / Total Predictions

Precision

Precision measures how many predicted positive purchases were actually correct.

Recall

Recall measures how many actual purchase cases were correctly identified.

F1 Score

F1 Score provides a balance between Precision and Recall and is considered one of the most important metrics for classification problems.

These evaluation metrics helped compare:

- Logistic Regression
- KNN
- Decision Tree
- SVM

and identify the best-fit model for the Online Purchase Prediction System.

8. Feature Scaling

Feature scaling is the process of transforming features into similar numerical ranges so that machine learning algorithms can perform efficiently.

In this project, StandardScaler was used for feature scaling.

Feature scaling was especially important for:

- KNN
- Logistic Regression
- SVM

because these algorithms depend on:

- distance calculations
- gradient optimization
- feature magnitude comparisons

Without feature scaling:

- features with larger numerical values dominate smaller features
- model performance decreases

After scaling, the models achieved improved Accuracy and F1 Score.

• PLATFORM REQUIRED:

Jupyter Notebook

• PROMPT USED TO GENERATE THE DATASET:

generate a dataset with 1,00,000 rows for an online purchase prediction system. the dataset should include the following columns user id (for example : User001), browsing time (min) , number of clicks, pages visited (integer) , add to cart count (integer), product category viewed, wishlist add (categorical values), bounce rate (min), exit pages (categorical value), time per page (min), user type, location, search keywords, session duration (min), product views (int), Purchase (0/1)

1 → Customer made a purchase

0 → Customer did not purchase.

ensure the dataset reflects realistic environments:

1. random noises in sensor readings,
2. missing (null) values in some entries (replace the blank cells by NaN),
3. outliers (extreme or abnormal values),
4. slight inconsistencies or variability in measurements

and merge two adjacent cells for every entry of every column in the excel sheet and round off the decimal values upto two decimal places.

• Description of Features Used in the Dataset:

The dataset used in this project consists of multiple features that capture user behavior, product interaction, and session characteristics. These features help in understanding customer intent and predicting whether a purchase will occur.

1. User Behaviour Features:

- Browsing Time:
Represents the total time a user spends on the website during all interactions. Higher browsing time may indicate greater interest in products.
- Session Duration:
Refers to the total time spent by a user in a single visit or session. Longer sessions often suggest deeper engagement with the platform.
- Number of Clicks:
Indicates the total number of clicks made by the user during a session. A higher number of clicks may reflect active exploration of products.
- Pages Visited:
Denotes the number of different pages viewed by the user. More page visits typically imply increased interest and information-seeking behavior.

2. Product Interaction Features:

- **Product Views:**
Represents the number of product pages viewed by the user. Frequent product views suggest a higher likelihood of purchase intent.
- **Add to Cart Count:**
Indicates how many items a user has added to their cart. This is a strong indicator of purchase intention, even if the purchase is not completed.
- **Product Category Viewed:**
Specifies the type or category of products browsed by the user. This helps in understanding user preferences and interests.
- **Wishlist Adds:**
Represents the number of products added to the wishlist. It indicates future purchase intent or interest in specific items.

3. Session Features:

- **Bounce Rate:**
Indicates whether a user leaves the website after viewing only one page. A high bounce rate generally suggests low engagement.
- **Exit Pages:**
Refers to the last page visited by the user before leaving the website. It helps identify where users drop off in their browsing journey.
- **Time per Page:**
Represents the average time spent by the user on each page. Higher values indicate deeper engagement with the content.

4. User Information (Optional):

- **User Type:**
Classifies users as new or returning visitors. Returning users are often more likely to make a purchase due to familiarity with the platform.
- **Location:**
Indicates the geographical region or country of the user. It can influence purchasing behavior based on demographics and accessibility.
- **Device Type:**
Specifies whether the user is accessing the platform via mobile, desktop, or tablet. User behavior may vary depending on device type.

5. Traffic Source Features:

- **Traffic Type:**
Identifies how the user reached the website, such as direct access, referral links, or social media. Different sources can influence user intent.
- **Search Keywords:**
Represents the keywords used by users to reach the website. It provides insight into user needs and search intent.

6. Target Variable:

- **Purchase (0/1):**
This is the output variable of the model.
 - **1:** Indicates that the user made a purchase
 - **0:** Indicates that the user did not make a purchase

This variable is used to train the classification model to predict future user behaviour

- **Books and Link Sources:**

1. Fundamental of AI and ML Textbook
2. Scikit Learn Documentation: scikit-learn.org
3. Data Preprocessing and Feature Engineering for Data Mining: www.researchgate.net
4. GeeksforGeeks (for reading up various topics of machine learning) :
www.geeksforgeeks.org